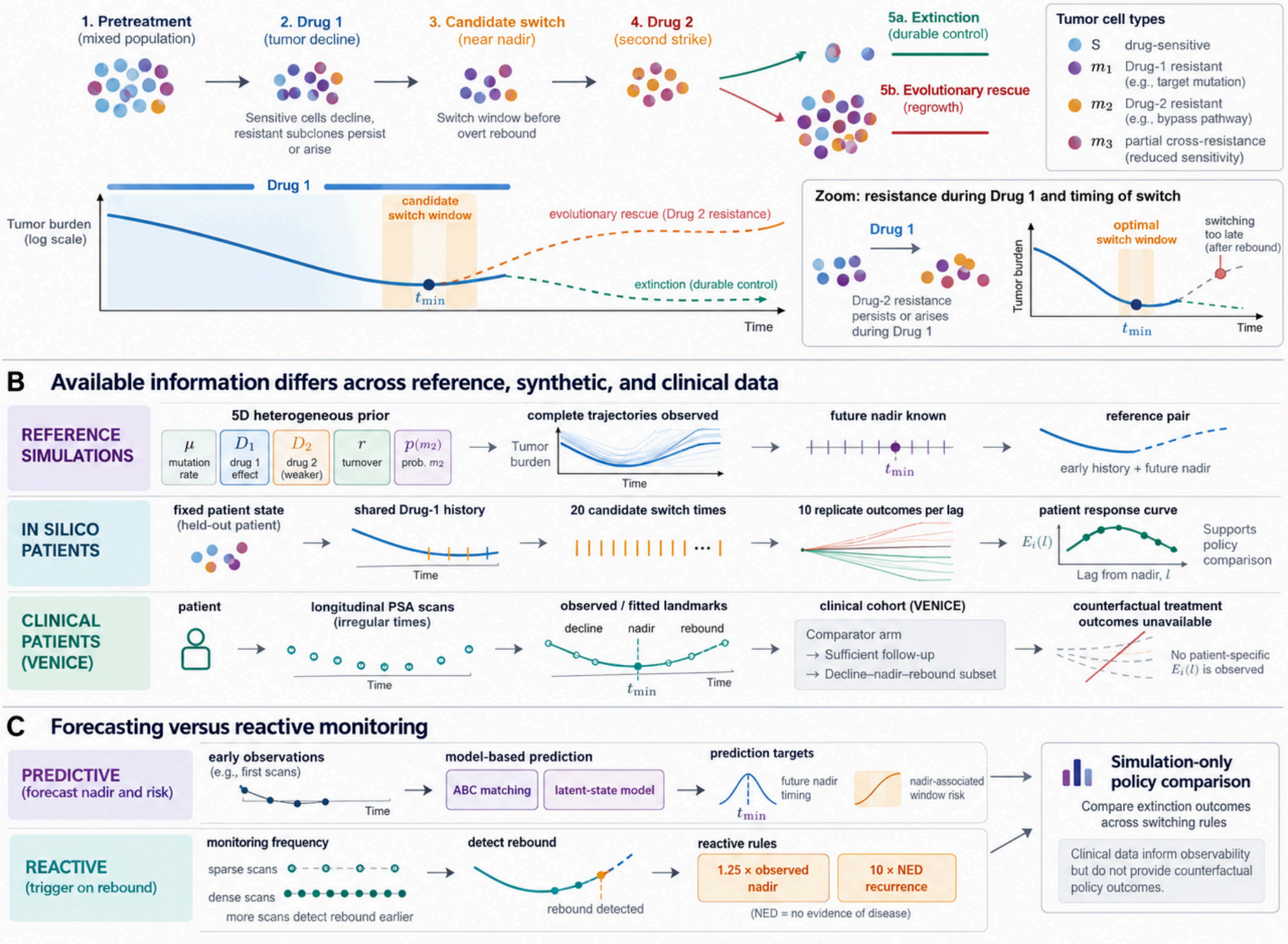

A
Evolutionary-rescue setting and candidate switching window
1. Pretreatment
(mixed population)
2. Drug 1
(tumor decline)
Sensitive cells decline, resistant subclones persist or arise
3. Candidate switch
(near nadir)
Switch window before overt rebound
4. Drug 2
(second strike)
5a. Extinction
(durable control)
5b. Evolutionary rescue
(regrowth)
Tumor cell types
S drug-sensitive
m_1 Drug-1 resistant (e.g., target mutation)
m_2 Drug-2 resistant (e.g., bypass pathway)
m_3 partial cross-resistance (reduced sensitivity)
Drug 1
candidate switch window
evolutionary rescue (Drug 2 resistance)
extinction (durable control)
Tumor burden (log scale)
t_{min}
Time
Zoom: resistance during Drug 1 and timing of switch
Drug 1
Drug-2 resistance persists or arises during Drug 1
optimal switch window
switching too late (after rebound)
Tumor burden
t_{min}
Time
B
Available information differs across reference, synthetic, and clinical data
REFERENCE SIMULATIONS
5D heterogeneous prior
μ mutation rate
D_1 drug 1 effect
D_2 drug 2 (weaker)
r turnover
$p(m_2)$ prob. m_2
complete trajectories observed
Tumor burden
Time
future nadir known
t_{min}
reference pair
early history + future nadir
IN SILICO PATIENTS
fixed patient state
(held-out patient)
shared Drug-1 history
Time
20 candidate switch times
10 replicate outcomes per lag
patient response curve
$E_i(l)$
Lag from nadir, l
Supports policy comparison
CLINICAL PATIENTS (VENICE)
patient
longitudinal PSA scans
(irregular times)
Time
observed / fitted landmarks
decline
nadir
rebound
t_{min}
Time
clinical cohort (VENICE)
Comparator arm
→ Sufficient follow-up
→ Decline–nadir–rebound subset
counterfactual treatment outcomes unavailable
No patient-specific $E_i(l)$ is observed
C
Forecasting versus reactive monitoring
PREDICTIVE
(forecast nadir and risk)
early observations
(e.g., first scans)
Time
model-based prediction
ABC matching
latent-state model
prediction targets
t_{min}
future nadir timing
nadir-associated window risk
REACTIVE
(trigger on rebound)
monitoring frequency
sparse scans
dense scans
more scans detect rebound earlier
detect rebound
rebound detected
reactive rules
1.25 × observed nadir
10 × NED recurrence
(NED = no evidence of disease)
Simulation-only policy comparison
Compare extinction outcomes across switching rules
Clinical data inform observability but do not provide counterfactual policy outcomes.